AUTOM-05: Dynatrace Workflows

> **Series:** AUTOM — Dynatrace Automation | **Notebook:** 5 of 9 | **Created:** January 2026 | **Last Updated:** 11/11/2026

Dynatrace Workflows is a built-in automation engine that enables event-driven actions directly within the platform. Unlike external tools, workflows run inside Dynatrace with full access to observability data.

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Prerequisites

Before starting this notebook, ensure you have:

Requirement	Description
Dynatrace SaaS	Tenant with Workflows enabled

Requirement	Description
Permissions	`automation:workflows:read` to view workflows; `automation:workflows:write`, `automation:workflows:run` and `app-engine:functions:run` to build and run them (Workflows security (DT docs))
Basic DQL	Understanding of Dynatrace Query Language

Learning Objectives

By the end of this notebook, you will:

- Understand Dynatrace Workflows architecture
- Know how to create event-driven automations
- Be able to implement auto-remediation patterns
- Connect workflows to external systems

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1. Introduction

Why Workflows?

Benefit	Description
Native Integration	Direct access to all Dynatrace data
No Infrastructure	No external systems to manage
Event-Driven	React to problems, alerts, schedules
Context-Aware	Full Dynatrace Intelligence context available
Secure	Credentials stored in Dynatrace vault

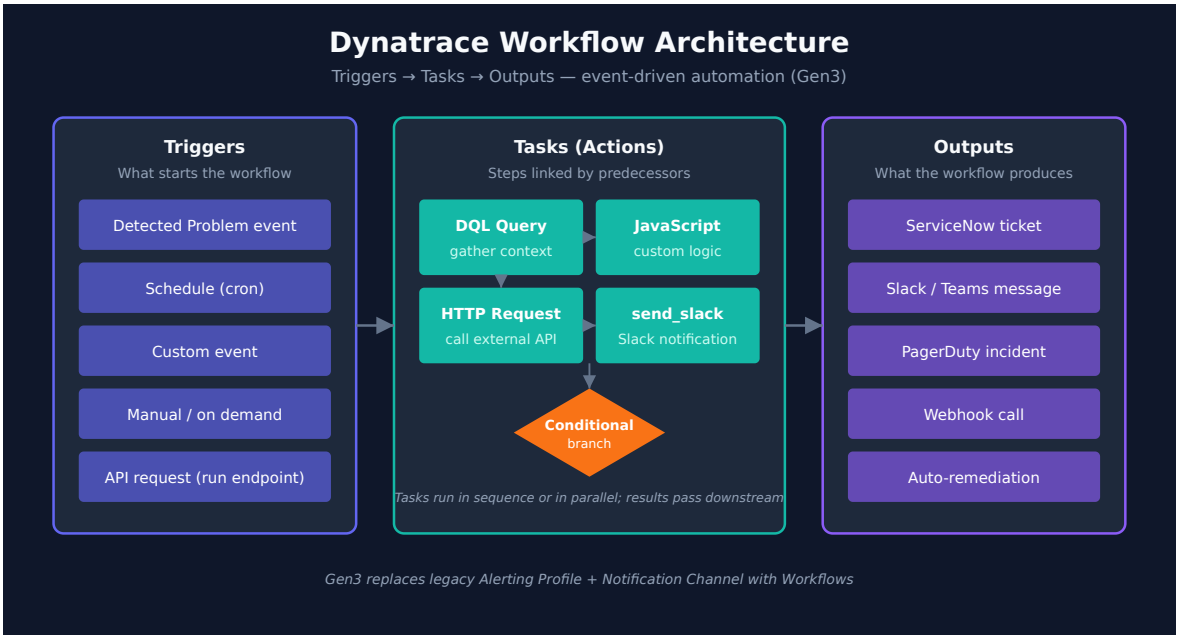
Common Use Cases

Use Case	Description
Notifications	Send alerts to Slack, Teams, email
Auto-Remediation	Restart services, scale resources
Ticketing	Create ServiceNow, Jira tickets
Enrichment	Add context to problems
Reporting	Generate scheduled reports

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2. Workflow Components

Workflow Architecture



Triggers

"A trigger can be a schedule, an event, a manual interaction (on demand), or an API request." There is no inbound-webhook trigger; an external system starts a workflow by calling its run endpoint (Build workflows (DT docs)).

Trigger Type	Description	Example
Problem	detected problem event	Availability issue detected
Event	Custom or ingest event	Deployment completed
Schedule	Time-based (cron)	Daily at 9 AM
Manual	User-initiated	On-demand execution
API request	Call to the workflow's run endpoint (API or SDK)	External system trigger

Tasks

Task Type	Description
HTTP Request	Call external APIs
Run JavaScript	Custom logic
DQL Query	Query Grail data
Send Notification	Slack, Teams, email
Create Issue	Jira, ServiceNow
Run Workflow	Call other workflows

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3. Creating Workflows

Access Workflows

Navigate to: **Apps** → **Workflows**

Or via URL:
`https://{tenant}.apps.dynatrace.com/ui/apps/dynatrace.automations/workflows`

Simple Notification Workflow

The examples in this notebook use the workflow **template** format (`schemaVersion: 3`) that the Workflows app imports and exports, trimmed for reading: the `metadata` block and

each task's `position` are left out, and connection fields are empty — select the connection in the editor after import.

> **Action IDs** come from Dynatrace's own workflow samples: `dynatrace.slack:slack-send-message`, `dynatrace.servicenow:snow-create-incident` and `dynatrace.email:send-email`, plus the core `dynatrace.automations:execute-dql-query`, `dynatrace.automations:run-javascript` and `dynatrace.automations:http-function`. Connector actions need the connector installed from Dynatrace Hub. If the editor shows a different ID for an action, use the one it shows. (Dynatrace-workflow-samples AGENTS.md (Dynatrace GitHub))

```
workflow:
  title: Problem Notification
  description: Send a Slack message for availability and error problems
  tasks:
    send_slack:
      name: send_slack
      description: Send Slack alert
      action: dynatrace.slack:slack-send-message
      input:
        connection: ""          # select the Slack connection in the editor
        channel: "&lt;channel-id&gt;" # Slack channel ID
        message: |-
          :warning: *Problem Detected*
          *Problem:* {{ event()["display_id"] }} — {{ event()["event.name"] }}
          *Category:* {{ event()["event.category"] }}
          *Status:* {{ event()["event.status"] }}
      predecessors: []
  trigger:
    eventTrigger:
      isActive: true
      triggerConfiguration:
        type: davis-problem
        value:
          categories:
            availability: true
            error: true
  schemaVersion: 3
```

Slack calls leave the platform, so allow `slack.com` under **Settings → General → External requests** before the first run.

Problem-trigger event fields are the Grail problem-record field names (`event.name`, `event.category`, `event.status`, `display_id`, `root_cause.smartscape_entity`, `smartscape.affected_entities`) — the same fields `fetch dt.davis.problems` returns — not the Classic problem-API names (`title`, `severity`, `impactLevel`, `rootCauseEntity`). The older `root_cause_entity_id`, `root_cause_entity_name` and `affected_entity_ids` still exist but are deprecated: *"root_cause_entity_id and root_cause_entity_name are deprecated in favor of root_cause.smartscape_entity , and affected_entity_ids in favor of smartscape.affected_entities . Build new payloads against the Smartscape fields."* Most problems carry no root cause, so read it with `.get()` as the examples below do.

Problem Trigger Configuration

Option	Description
Problem state	Active (default), active or closed, or closed only
Event category	Availability, Error, Slowdown, Resource, Custom, Info, Monitoring unavailable (`resource` matches `event.category == "RESOURCE_CONTENTION"`)
Severity	The level at or above which problems start the workflow
Affected entities	Filter by entity tags — all defined tags, or any of them
Minimum duration	Postpone the trigger until the problem has been open 5 minutes to one week
Wait for root cause analysis	Start only after root-cause analysis has finished
Additional custom filter query	A DQL matcher, for example `maintenance.is_under_maintenance == false`

There is no management-zone or entity-type option: scope with entity tags or affected entities. Do not scope on the root cause — the alert-notification upgrade guide: *"Do not filter on root_cause.smartscape_entity."* … *"A root-cause condition silently drops those problems. Filter on tags or affected entities instead."* (Upgrade guide — alerting and notifications (DT docs), Event triggers (DT docs), dynatrace_automation_workflow resource (Dynatrace GitHub), ServiceNow incident sample (Dynatrace GitHub))

Using JavaScript Tasks

Custom logic with JavaScript:

```
// Task: Enrich problem data
import { execution } from '@dynatrace-sdk/automation-utils';

export default async function () {
  // The triggering event is on the execution, not a global
  const ex = await execution();
  const problem = ex.event() ?? {};

  // Build enrichment data
  return {
    problemId: problem['display_id'],
    name: problem['event.name'],
    affectedEntities: (problem['smartscape.affected_entities'] ??
problem['affected_entity_ids'] ?? []).length,
    rootCauseEntity: problem['root_cause.smartscape_entity']?.name ||
'Unknown',
    timestamp: new Date().toISOString()
  };
}
```

A predecessor task's output is `await ex.result('task_name')` (or the standalone `result('task_name')` export) — see the automation-utils SDK reference (Dynatrace Developer). List that task under `predecessors`: *"A task can reference only the outputs of its direct predecessors in the workflow graph. If the producing task is not a direct predecessor, the expression resolves to Undefined variables at runtime."* A task with no predecessors starts as soon as the workflow does. (Run JavaScript action (DT docs))

Using DQL Tasks

Query data within workflows:

```
tasks:
  get_metrics:
    name: Get CPU Metrics
    action: dynatrace.automations:execute-dql-query
    input:
      query: |
        timeseries cpu = avg(dt.host.cpu.usage), from:-30m, filter:
{dt.smartscape.host == toSmartscapeId("{{
(event().get('root_cause.smartscape_entity') or {}).get('id', '') }}")}
```

4. Actions and Integrations

Built-in Connectors

Connector	Actions
Slack	Send message, post to channel
Microsoft Teams	Send adaptive card
Jira	Create/update issue
ServiceNow	Create, search and comment on incidents; import ServiceNow groups as Ownership teams
PagerDuty	Create incident
Jenkins	Trigger builds and query build status
Email	Send email notification

Connectors other than Email are installed from Dynatrace Hub. (Workflow actions (DT docs), ServiceNow Connector (DT docs))

HTTP Request Task

Call any REST API:

```
tasks:
  call_api:
    name: Call External API
    action: dynatrace.automations:http-function
    input:
      url: "https://api.example.com/incidents"
      method: POST
      headers:
        Content-Type: application/json
        # No Authorization header: select a Credential Vault token in the
        # action's Authentication input instead (see Credential Management
below)
      payload: |
        {
          "title": "{{ event()['event.name'] }}",
          "category": "{{ event()['event.category'] }}",
          "source": "dynatrace"
        }
```

The request body goes in `payload` (*"The payload of the HTTP request. Set an appropriate content-type header."*); `body` is a field of the action's *result*, not an input. Add the target domain under **Settings → General → External requests** first: *"All HTTP calls are validated against the global allowlist."* (HTTP Request action (DT docs))

Credential Management

Store credentials securely:

1. Go to **Settings → Credential Vault**
2. Create a new credential (Token or Basic)
3. In an HTTP Request task, select it in the **Authentication** input: *"The HTTP Request action supports using credentials from the credential vault for Basic and Token authentication."* In a Run JavaScript task, read it with `credentialVaultClient`.

The workflow expression language has no `env` or `credential` object, so `{{ env.API_TOKEN }}` and `{{ credential.MY_API_KEY }}` are not valid references. Do not fall back to a literal header either — header values show in the workflow monitor, and Dynatrace advises: *"We strongly advise you not to expose any secret, but instead to use the authentication configuration via Credential Vault."* (HTTP Request action (DT docs), Jinja expressions for Workflows (DT docs), Run JavaScript action (DT docs))

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5. Auto-Remediation Patterns

Pattern: Restart Service

```
workflow:
  title: Auto-Restart on Crash
  tasks:
    check_entity:
      name: check_entity
      description: Validate the root-cause entity type
      action: dynatrace.automations:run-javascript
      input:
        script: |-
          import { execution } from '@dynatrace-sdk/automation-utils';
          export default async function () {
            const ex = await execution();
            const rootCause = (ex.event() ?? {})
['root_cause.smartscape_entity'] ?? {};
            if (rootCause.type !== 'PROCESS') {
              return { skip: true };
            }
            return { entityId: rootCause.id, skip: false };
          }
      predecessors: []
    restart_service:
      name: restart_service
      description: Call the restart runbook
      action: dynatrace.automations:http-function
      input:
        url: "https://runbook.example.com/restart"
        method: POST
        headers:
          Content-Type: application/json
        payload: |-
          { "entityId": "{{ result('check_entity').entityId }}" }
      predecessors:
        - check_entity
      conditions:
        states:
          check_entity: OK
        custom: "{{ result('check_entity').skip == false }}"
        else: SKIP
  trigger:
    eventTrigger:
      isActive: true
```

```
      triggerConfiguration:
        type: davis-problem
        value:
          categories:
            availability: true
  schemaVersion: 3
```

Pattern: Scale on High CPU

```
workflow:
  title: Auto-Scale on Resource Pressure
  tasks:
    check_cpu:
      name: check_cpu
      description: Query current CPU of the root-cause host
      action: dynatrace.automations:execute-dql-query
      input:
        query: |-
          timeseries cpu = avg(dt.host.cpu.usage), from:-30m, filter:
{dt.smartscape.host == toSmartscapeId("{{
(event().get('root_cause.smartscape_entity') or {}).get('id', '') }}")}
          | fieldsAdd avg_cpu = arrayAvg(cpu)
      predecessors: []
    scale_up:
      name: scale_up
      description: Trigger scale event
      action: dynatrace.automations:http-function
      input:
        url: "https://api.cloud.example.com/scale"
        method: POST
      predecessors:
        - check_cpu
      conditions:
        states:
          check_cpu: OK
        custom: "{{ result('check_cpu').records | length &gt; 0 and
result('check_cpu').records[0].avg_cpu &gt; 90 }}"
        else: SKIP
  trigger:
    eventTrigger:
      isActive: true
      triggerConfiguration:
        type: davis-problem
        value:
          categories:
            resource: true
  schemaVersion: 3
```

This pattern only fits problems whose root cause is a host. On a validation tenant over seven days (10/02/2026), 16 resource-contention problems had a host as root cause, 45 a process group, and 1,477 no root-cause entity at all; for anything but a host the query returns no records, and `records[0].avg_cpu` would be undefined. The trigger deliberately has no root-cause filter — *"A root-cause condition silently drops those problems."* — so

the query does the narrowing: it filters on the root-cause host, returns no records for any other root cause or none, and the `length > 0` guard then skips the scale call. A `PROCESS` root cause in the restart pattern carries a Smartscape ID (`PROCESS-…`), not the classic `PROCESS_GROUP_INSTANCE-…`, so point the runbook at that ID. (Upgrade guide — alerting and notifications (DT docs))

> A bare `timeseries avg(dt.host.cpu.usage) | filter dt.entity.host == …` fails with `FIELD_DOES_NOT_EXIST`: without `by:{dt.entity.host}` the series carries no host dimension to filter on. Filter inside `timeseries` (as above) or add the `by:` clause. Compare `dt.smartscape.host` through `toSmartscapeId(…)`: a `smartscape_id` never equals a plain string.

Pattern: Create Incident Ticket

```
workflow:
  title: ServiceNow Incident Creation
  tasks:
    create_incident:
      name: create_incident
      description: Create a ServiceNow incident
      action: dynatrace.servicenow:snow-create-incident
      input:
        connectionId: ""        # select the ServiceNow connection in the
editor
        shortDescription: "Dynatrace: {{ event()['event.name'] }}"
        description: |-
          Problem detected by Dynatrace Intelligence

          Problem: {{ event()['display_id'] }}
          Name: {{ event()['event.name'] }}
          Category: {{ event()['event.category'] }}
          Root Cause: {{ (event().get('root_cause.smartscape_entity') or
{}).get('name', 'not yet determined') }}
        urgency: "2"
        impact: "2"
        correlationId: DT_{{ event()['event.id'] }}
      predecessors: []
    record_ticket:
      name: record_ticket
      description: Keep the incident reference for later tasks
      action: dynatrace.automations:run-javascript
      input:
        script: |-
          import { result } from '@dynatrace-sdk/automation-utils';
          export default async function () {
            const incident = await result('create_incident');
            return { incidentUrl: incident?.url, status: 'created' };
          }
      predecessors:
        - create_incident
      conditions:
        states:
          create_incident: OK
  trigger:
    eventTrigger:
      isActive: true
      triggerConfiguration:
        type: davis-problem
        value:
```

```
          categories:
            availability: true
            error: true
  schemaVersion: 3
```

Input names follow Dynatrace's ServiceNow sample, which also searches for an existing incident by `correlationId` (`snow-search-incidents`) before creating one, so a re-triggered problem does not open a second ticket. (ServiceNow incident sample (Dynatrace GitHub))

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6. Best Practices

Workflow Design

Practice	Description
Single responsibility	One workflow, one purpose
Idempotency	Safe to run multiple times
Error handling	Handle failures gracefully
Logging	Add descriptive task names
Testing	Use manual triggers for testing

Trigger Configuration

Practice	Description
Scope narrowly	Use the trigger's entity-tag filter and a custom filter query
Avoid duplicates	Check if action already taken
Debounce	Use the trigger's **Minimum duration** option (5 minutes to one week) for flapping problems

Security

Practice	Description
Credential vault	Never hardcode secrets
Least privilege	Minimal API permissions
Audit trail	Log all remediation actions

Error Handling

Workflows have no `on_error` or `error()` construct. Retries are a task property (`count` 1–99, `delay` in **seconds** 1–3600), and error routing is a downstream task whose **condition** is the upstream task's state (`SUCCESS` , `ERROR` , `ANY` , `OK` , `NOK`):

```
tasks:
  api_call:
    name: Call API
    action: dynatrace.automations:http-function
    retry:
      count: 3
      delay: 30          # seconds

  error_handler:
    name: error_handler
    action: dynatrace.slack:slack-send-message
    predecessors:
      - api_call
    conditions:
      states:
        api_call: ERROR
    input:
      connection: ""          # select the Slack connection in the editor
      channel: "&lt;channel-id&gt;"
      message: "Task api_call failed after 3 retries"
```

The same fields in Terraform form are documented on the dynatrace_automation_workflow resource (Dynatrace GitHub) — `retry { count, delay, failed_loop_iterations_only }` and `conditions { states = { … } }` .

7. Recent Enhancements

New Workflow Features (2025-2026)

Dynatrace has added several capabilities to the Workflows engine:

Feature	Description
Sub-workflows	Call reusable workflows from within other workflows for modular automation
Approval Requests	Built-in human approval gates for governed automation
Workflow Drafts	Save work-in-progress workflows without deploying them
Simple Workflows	Quickly automate single-step tasks (e.g., send a Slack notification) at no additional cost
Persistent Execution Data	Retain execution history for analysis and refinement
Real-time Notifications	Get notified when workflows complete or fail
ServiceNow Connector	Create, search and comment on incidents; import ServiceNow groups as Ownership teams

Agentic Workflows

An agentic workflow is a workflow with at least one agentic action — a task that reasons over live environment data instead of producing a fixed output:

Capability	Description
Prompt generative AI	Agentic action for generative tasks, such as a summary
Prompt agentic AI	Agentic action that calls tools to query and act on your environment
Request Approval	A task that pauses for a human decision before a consequential step

Dynatrace's guidance: *"When an agentic workflow drives consequential actions, consider adding a Request Approval task to keep a human in the loop."* Agentic actions are tasks like any other, so the trigger, predecessor and condition patterns above apply unchanged. (Workflows concepts (DT docs))

8. Next Steps

Workflows vs External Automation

Scenario	Tool Choice
Event-driven from Dynatrace	Workflows
Config deployment	Monaco/Terraform
Complex orchestration	External (Ansible, etc.)
Cross-platform actions	Workflows with HTTP tasks

Continue the Series

Next Notebook	Focus
AUTOM-06: Dynatrace SDKs	Programmatic access with TypeScript/Python

Additional Resources

- Workflows Documentation
- Workflow Actions Reference

Summary

In this notebook, you learned:

- Dynatrace Workflows architecture and components
- Creating event-driven automations with triggers and tasks
- Implementing auto-remediation patterns
- Best practices for workflow design and security

> **Key Takeaway:** Workflows provide native automation inside Dynatrace. Use them for event-driven actions like notifications, ticketing, and auto-remediation. For configuration management, continue using Monaco or Terraform.

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*Continue to **AUTOM-06: Dynatrace SDKs** to learn programmatic access patterns.*

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